MILITARY SPECIFICATION

LAMP, MERCURY XENON VAPOR, SEARCHLIGHT

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers mercury xenon vapor lamps used in searchlights.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.
MIL-I-45208 - Inspection System Requirements.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF TRANSPORTATION
MATERIALS TRANSPORTATION BUREAU
Hazardous Materials Regulations 49 CFR Parts 171-177.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402).

3. REQUIREMENTS

3.1 Qualification. Lamps furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 Materials. The materials shall be of the best commercial quality for the purpose intended. Reclaimed materials shall be used to the maximum extent possible.

3.3 Assembly. Lamps shall be provided with flexible leads and mounting collars and shall conform to the dimensions shown on figure 1. The mounting points shall be the two collars. Lamps shall be held in the searchlight by means of clamps which engage the collars. Collars shall press against stops (not shown on figure) which will provide the correct vertical position.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:55Z
Check the source to verify that this is the current version before use.

3.4 Electrical characteristics. Electrical characteristics shall be as follows:

(a) Power supply - 115 volts + 5 percent regulation, 60 hertz (Hz) ± 5 percent.
(b) Lamp power input - 850 + 100 watts, when operated in the circuit shown on figure 2.
(c) Lamp voltage - 65 volts ± 5 volts.

3.5 Arc characteristics. Arc characteristics of the lamp shall be as follows when operated at 1000 watts, plus or minus 2 percent.

(a) Ignition time - 5 seconds maximum.
(b) Arc length - 6.5 ± 0.5 minimum.
(c) Arc width - 4.8 mm minimum measured across center of arc, between the two points having 10 percent of the luminance of the center of the arc.
(d) Light output:
(1) At equilibrium (new lamp) 50,000 lumens minimum.
(2) At equilibrium (after 500 hours operation) 15,000 lumens minimum.

3.6 Warm-up characteristics. Lamps shall start and provide a luminous flux of at least 13,000 lumens 2 minutes after ignition when tested in accordance with 4.5.2.

3.7 Operation. Lamps shall give no indication of electrical or mechanical defects when tested in accordance with 4.5.3. The lamp shall be capable of intermittent operation in all positions of searchlight elevation up to 90 degrees. The light center position tolerances shall be as follows:

(a) Vertical - plus or minus 1/20 inch.
(b) Transverse to searchlight axis - plus or minus 1/20 inch.
(c) A long searchlight axis focusing direction along line connecting arc and bulb evacuation tip - plus or minus 1/10 inch.

3.8 Bulb enclosure. Lamps shall be furnished with a metal enclosure completely enclosing the bulb. The metal enclosure shall be cylindrical in form, split vertically, and hinged on one side to open and close in the manner of a clam shell. A mechanical locking device shall be provided for locking the enclosure in the closed position. It shall be capable of being easily detached from the lamp in the open position.

3.9 Workmanship. The bulb shall be free of imperfections such as bubbles, scratches, and visible distortions which could impair the proper functioning of the lamp. The metal enclosure shall be smooth and free from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208 (see 6.2.2).

4.2 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center (NAVSEC). Qualification inspection shall consist of the examination and tests shown in table I.

TABLE I. Qualification inspection and quality conformance inspection.

Test	Requirement paragraph	Test paragraph	Qualification	Quality Conformance	
				Group A	Group B
Surface examination	3.9	4.4	X	X	
Arc characteristics	3.5	4.5.1	X		X
Warm-up	3.6	4.5.2	X		X
Operation	3.7	4.5.3	X	X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:55Z
Check the source to verify that this is the current version before use.

4.2.1 Qualification samples. Three lamps shall be subjected to and shall pass the qualification inspection of 4.2.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of the group A and B inspections (see 4.3.3 and 4.3.4).

4.3.1 Inspection lot. For purposes of sampling, a lot shall consist of not more than 500 lamps offered for delivery at one time.

4.3.2 Sampling.

4.3.2.1 Sampling for group A inspection. A random sample of lamps shall be selected from each inspection lot in accordance with inspection level II of MIL-STD-105 for the group A inspection (see 4.3.3). The Acceptable Quality Level (AQL) shall be 1.0 percent defective. Any lamp in the sample containing one or more visual or dimensional defects or failing the 5-hour operating test shall be rejected, and if the number of defective lamps in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3.2.2 Sampling for group B inspection. A random sample of lamps shall be selected from each inspection lot in accordance with level S-1 of MIL-STD-105 for the group B inspection of 4.3.4. If any lamp fails any group B test, the lot represented by the sample shall be rejected.

4.3.3 Group A. The sample lamps selected in accordance with 4.3.2.1 shall be subjected to the group A tests shown in table I.

4.3.4 Group B. The sample lamps selected in accordance with 4.3.2.2 shall be subjected to the group B tests shown in table I.

4.3.5 Quality conformance inspection reports. Quality conformance inspection reports shall be prepared as specified (see 6.2.2).

4.4 Surface examination. Each of the sample lamps shall be visually and dimensionally examined to verify compliance with this specification.

4.5 Test procedures.

4.5.1 Arc characteristics. The lamps shall be tested for compliance with 3.5, when operated in the circuit shown on figure 2. For the 500-hour test (see 4.5.3), the operating cycle shall be 2 hours per start.

4.5.2 Warm-up. The lamps shall be tested for compliance with 3.6, when operated in the circuit shown on figure 2.

4.5.3 Operation. After the lamps have been visually and dimensionally examined they shall be tested for compliance with 3.7, when operated in the circuit shown on figure 2. The length of test shall be as follows:

(a) Qualification inspection: 500 hours.
(b) Quality conformance inspection: 5 hours.

4.6 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation-packaging and packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

5.1 Preservation-packaging, packing, and marking. Lamps shall individually preserved-packaged level A or C, packed level A, B, or C as specified (see 6.2.1) and marked in accordance with MIL-E-17555 and as specified herein.

5.1.1 Items containing metallic mercury or mercury compounds. The preservation-packaging and packing requirements specified in 5.1 for the level required, shall comply with the Department of Transportation (DOT), Materials Transportation Bureau, Hazardous Materials Regulations, 49 CFR Parts 171-177.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:55Z
Check the source to verify that this is the current version before use.

5.1.2 Cushioning, dunnage, and wrapping materials.

5.1.2.1 Level A preservation-packaging and levels A and B packing. Use of all types of loose-fill materials for packaging and packing applications such as cushioning, filler or dunnage is prohibited for materials destined for shipboard installation/stowage.

5.1.2.2 Level C preservation-packaging and packing. When loose fill type materials are used for packaging and packing applications such as cushioning, filler and dunnage, all containers (unit, intermediate and shipping) shall be marked or labelled with the following information:

"CAUTION

Contents cushioned etc., with loose-fill material. Not to be taken aboard ship. Remove and discard loose-fill material before shipboard stowage. If required, recushion with cellulosic material, bound fiber, fiberboard or transparent flexible cellular material."

5.1.3 Special marking.

5.1.3.1 Items containing metallic mercury or mercury compounds. Whenever an item containing mercury is enclosed in a container (interior package or shipping) which covers or hides the item identifying plate or label or when the item size causes identification to be impractical, each container shall be marked, tagged or labelled with the information specified hereinafter. Tags and labels shall be affixed to the outer surface of each container and coated with a transparent non-fading waterproof coating. In addition, all packages and containers (unit, intermediate, and shipping) shall be labelled with the DOT, Class B poisonous material label.

"CAUTION-MERCURY
USE APPROVED BY NAVSEA

Contains _____ Mercury or _____
mercury compound

in 0 _____
Name/location of part

In case of spillage in excess of two ounces by weight of mercury isolate area and notify ship's Commanding Officer and Industrial Hygienist, Medical Department, or equivalent."

6. NOTES

6.1 Intended use. The mercury xenon vapor lamps covered by this specification are intended for use in type II searchlights of MIL-S-19551.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preservation and packaging, packing, and marking requirements (see 5.1).

6.2.2 Data requirements. When this specification is used in a procurement which invokes the provision of the "Requirements for Data" of the Armed Services Procurement Regulations (ASPR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the ASPR are not invoked in a procurement, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

Paragraph	Data requirements	Applicable DID
4.1.1	Inspection system program plan	DI-R-4803
4.3.5	Test/inspection report	UDI-T-23473

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:55Z
Check the source to verify that this is the current version before use.

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the procuring/purchasing activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item procured to this specification. This does not apply to specific data which may be required for each procurement regardless of whether an identical item has been supplied previously (for example, test reports).

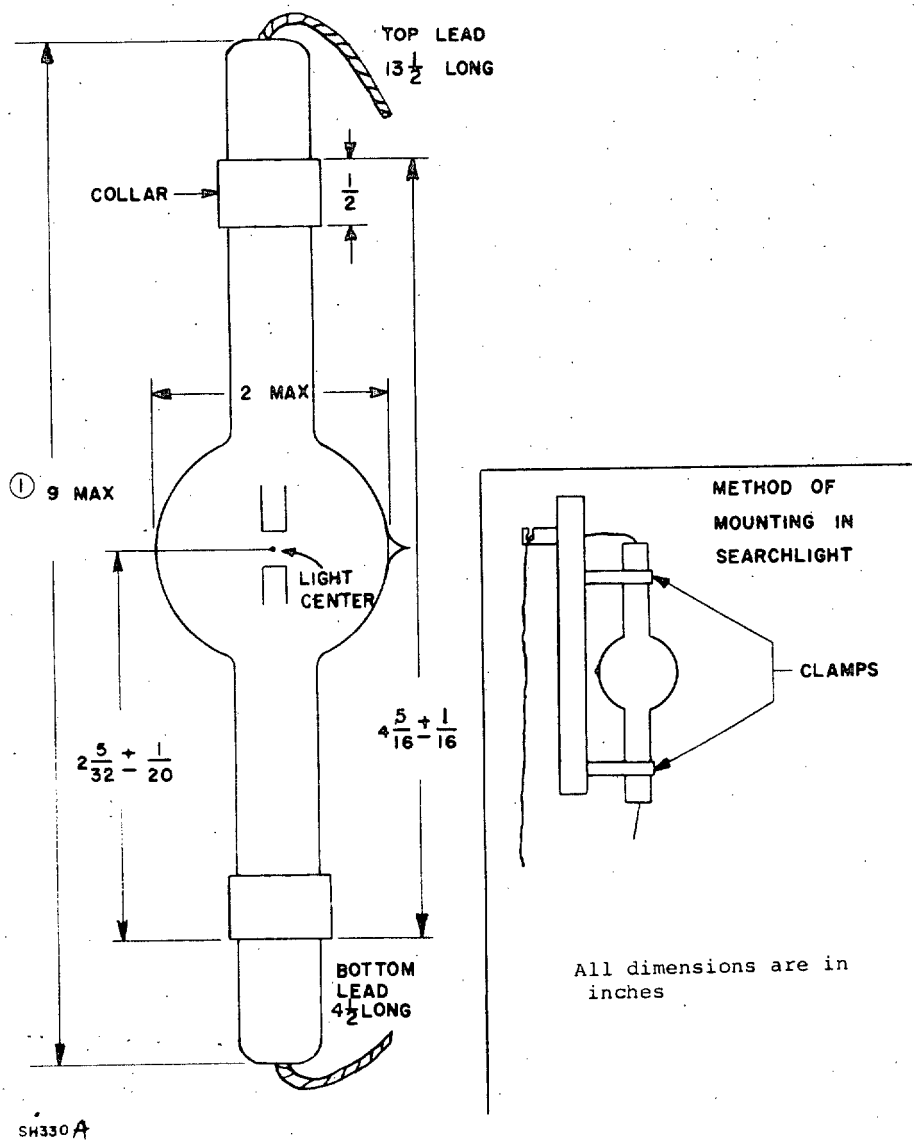
6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL-18052 whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Changes from previous issue. The symbol "#" is not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6240-N313)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:55Z
Check the source to verify that this is the current version before use.



NOTE: 1 Excluding leads

FIGURE 1. Mercury-xenon lamp.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:55Z
Check the source to verify that this is the current version before use.

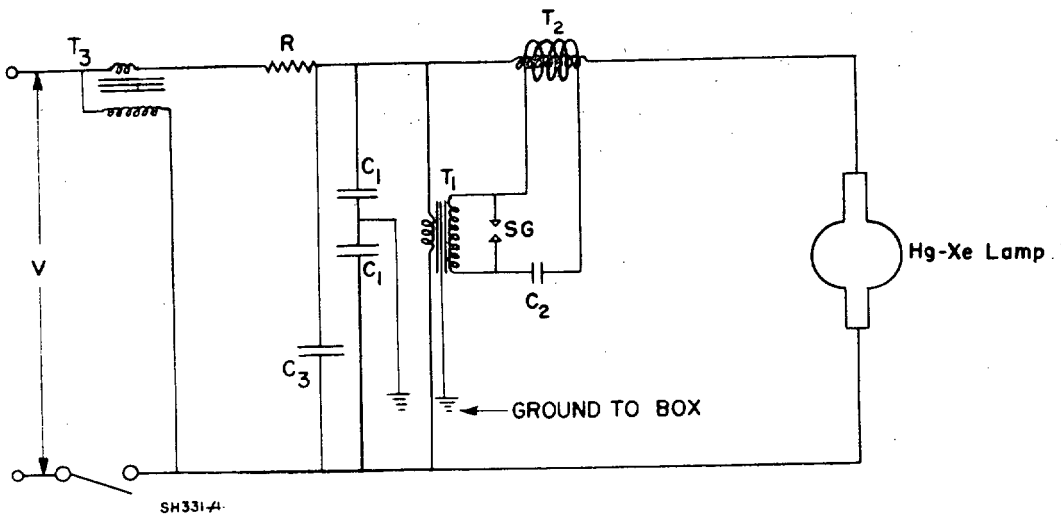


FIGURE 2. Starting circuit.

R = Ballast resistor, 4 nickel wire 6-1/4 to 7-1/4 ohms cold (27 ohms hot) each and one nichrome wire, 10.4 ohms in parallel.
T_1 = Transformer 115-volt, 60 Hz primary, 3800-volt open circuit secondary, short circuit current 20-25 milliampere (mA).
T_2 = High Frequency transformer, 3-turn primary, 45-turn secondary.
C_1 = By-pass condenser, two 0.0033 microfarad (µF), ceramic button, 1500-volt d.c.
C_2 = Condenser, four 0.0005 µF, ceramic, 20,000-volt in parallel.
C_3 = By-pass condenser, 2 µF -330-volts a.c. paper.
T_3 = Line booster transformer, 115-volt, 60 Hz primary, 130-volt secondary, 18 amperes.
V = Line voltage, 115 volts ± 5 percent.
SG = Spark gap, single open gap 0.048 inch spacing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:55Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-L-18052B 26 JUNE 1978

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:55Z
Check the source to verify that this is the current version before use.